Eliseo Bao

Ph.D. Student in Computer Science

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Research Interests

Natural Language Processing (NLP), with a focus on mental-health applications, explainability, and multilingual/underrepresented-language modeling, and the simulation and evaluation of LLM-based conversational agents.

Education

Ph.D. **University of A Coruña**, Computer Science — Galicia, Spain
- Programme site — 2024 – present

M.Sc. **University of A Coruña**, Computer Science — Galicia, Spain
Extraordinary Master's Award, 2023/2024 cohort. — 2024
- Programme site

B.Sc. **University of A Coruña**, Computer Science — Galicia, Spain
- Programme site — 2022

Tech. **IES San Clemente**, Application Development — Galicia, Spain
- Programme site — 2018

Experience

Universidade da Coruña, Predoctoral Fellow — 2024 – present
- Xunta de Galicia Predoctoral Grant ED481A-2024-079. — 2 years

Natural Language Processing Group, University of Sheffield, Visiting Researcher — Sheffield, UK
- Hosted by Dr. Xi Wang; work on LLM persona simulation led to the Eval4Sim paper (CIKM 2026). — Oct 2025 – Dec 2025 (3 months)

CITIC Centro de Investigacion TIC - Universidade da Coruña, Research Assistant — 2023 – 2024
- Working at the Information Retrieval Lab (IRLab) within the project PLEC2021-007662 Big-eRisk Early Prediction of Personal Risks on Massive Data funded by MCIN/AEI/10.13039/501100011033 and by the European Union NextGenerationEU/PRTR. — 1 year

ITG Instituto Tecnológico de Galicia, Internship — 2022 – 2023
- Contributed to the CEL.IA group, developing NLP solutions, with a focus on the ASR research line. — 1 year

NATICA, SISTEMAS DE INFORMACION SL, Apprenticeship — 2018 – 2018 (1 year)

Publications

Improving clinical reliability of LLM reasoning for depression assessment via structured generation and GRPO — July 2026
Shows GRPO-trained LLMs can explain depression-screening decisions through DSM-5-aligned reasoning without sacrificing classification accuracy.
Eliseo Bao, Anxo Pérez, Javier Parapar
10.1177/00368504261467483 (Science Progress, 109(3))

Eval4Sim: An Evaluation Framework for Persona Simulation 2026
Introduces a three-dimensional framework (adherence, consistency, naturalness) for evaluating how well LLM personas simulate human conversation.
Eliseo Bao, Anxo Pérez, Xi Wang, Javier Parapar
(Proc. of the 35th ACM Int'l Conference on Information and Knowledge Management (CIKM 2026). To appear)

Learning Evidence of Depression Symptoms via Prompt Induction July 2026
Proposes Symptom Induction, compressing labeled examples into interpretable guidelines that improve LLM classification of depressive symptoms.
Eliseo Bao, Anxo Pérez, David Otero, Javier Parapar
10.1145/3805712.3809873 (Proc. of the 49th International ACM SIGIR Conference (SIGIR 2026), Melbourne, Australia)

ReDSM5: A Reddit Dataset for DSM-5 Depression Detection Nov 2025
A 1,484-post Reddit dataset with sentence-level, clinician-annotated DSM-5 depressive-symptom labels and rationales.
Eliseo Bao, Anxo Pérez, Javier Parapar
10.1145/3746252.3761610 (Proc. of the 34th ACM Int'l Conference on Information and Knowledge Management (CIKM 2025), Seoul, Republic of Korea)

Towards Explainable and Safe Systems for Health Data July 2025
Doctoral consortium paper outlining a research plan for transparent, explainable, and safe health information systems.
Eliseo Bao
10.1145/3726302.3730121 (SIGIR 2025 Doctoral Consortium, Padua, Italy)

How does depression talk on social media? Modeling depression language with relevance-based statistical language models 2025
Builds relevance-based statistical language models of depression-related language to rank users by risk across Reddit and Twitter.
Eliseo Bao, Anxo Pérez, David Otero, Javier Parapar
10.1016/j.osnem.2025.100339 (Online Social Networks and Media, 50, 100339)

MindWell: A Conversational Agent for Professional Depression Screening on Social Media Apr 2025
Demonstrates an open-source RAG chatbot that helps clinicians review social media history for BDI-II depressive symptoms.
Eliseo Bao, Anxo Pérez, Javier Parapar
10.1007/978-3-031-88720-8_9 (Advances in Information Retrieval — ECIR 2025, LNCS, Part V, pp. 47-52, Lucca, Italy)

Explainable depression symptom detection in social media Sept 2024
Transformer-based models that both detect and explain depressive symptom markers in social media text.
Eliseo Bao, Anxo Pérez, Javier Parapar
10.1007/s13755-024-00303-9 (Health Information Science and Systems, 12(1), 47)

Adapting Large Language Models for Underrepresented Languages 2024
Introduces a Galician-adapted Llama 3.1 model and instruction dataset, outperforming the base model and another Galician model.
Eliseo Bao, Anxo Pérez, Javier Parapar
(VII Congreso XoveTIC, Universidade da Coruña — CICAS-Sngular Award)

MindWell: an open-source chatbot to assist in the detection and monitoring of depressive disorders 2024
Designed and evaluated the MindWell chatbot with a professional psychologist.
Eliseo Bao
(M.Sc. Thesis, Universidade da Coruña)

Ranking of Reddit users using Relevance Models for depressive disorders 2022
Built depression-risk vocabularies and ranking methods from Reddit using relevance-based language models.
Eliseo Bao
(B.Sc. Thesis, Universidade da Coruña)

Awards

CICAS-Sngular Award 2024
Awarded for the best work in line with best practices for Open Science, Open Data, Open Access and Open Source. Certification · Publication

Extraordinary Master's Award 2025
M.Sc. in Computer Engineering, 2023/2024 cohort.

B.Sc. thesis category award 2023
Certification · YouTube video

Finalist, Best Open Source University Project Award 2022
Publication

Open Source Projects

ReDSM5
Reddit dataset for DSM-5 depression detection.

Eval4Sim
Evaluation framework for LLM persona simulation.

Cabuxa
Llama 3.1 8B Instruct models and instruction datasets for Galician.

Talks & Outreach

- **2026** — "AI Against Hate," CITIC Day of Science in the Street.
- **2025** — MindWell demo presentation, ECIR 2025, Lucca, Italy.
- **2024** — "Adapting Large Language Models for Underrepresented Languages," VII Congreso XoveTIC, A Coruña.

Languages

Galician
Native

Spanish
Native

English
Professional working proficiency